



PRODUCTION OPTIMIZER

Well Decline + Recommendation Approval
Supervisor

Powered by Databricks Apps · Unity Catalog · AI/BI Genie ·
Mosaic AI

DIGITAL TWIN + GENIE + MULTI-AGENT APPROVAL SUPERVISOR FOR MATURE-FIELD PRODUCTION

Production Optimizer combines a 3D digital twin, governed natural-language access via AI/BI Genie, and a five-specialist Recommendation Approval Supervisor that fans out across Foundation Model API, UC SQL, and Vector Search to synthesise APPROVE / APPROVE-WITH-MODS / DEFER / REJECT for any optimization recommendation.

◆ BUSINESS VALUE

- **Catch decline early** with Arps fits + EUR + remaining-reserves per well
- **Approve high-impact recs faster** — five specialists run in parallel; verdict in seconds
- **Audit every decision** — every supervisor verdict logged with cited evidence
- **Self-service insight** — engineers ask Genie in plain English, not SQL
- **One governance plane** — Unity Catalog covers tables, models, queries, and audit

◆ CHALLENGES ADDRESSED

- **Decision drift** — engineers stitch 5+ tools per recommendation; nothing tied together
- **Manual interpretation** — decline curves and rec impact reviewed by hand
- **Slow approval cycles** — sign-off chains stretch days; capex bleeds
- **No precedent context** — past recommendations live in spreadsheets, not the workflow
- **Genie + Supervisor in one app** — chat and verdict surfaces unified, not two products

◆ WHO IT'S FOR



Production Engineer

Approves daily recs; chats Genie about per-well rates and history



Reservoir Engineer

Reviews decline curves + EUR + analog references via Supervisor



Asset Manager

Audits supervisor verdicts; defers high-risk recs to next maintenance window



Operations Team

Operates the digital twin; uses Genie for ad-hoc field questions

5

AI Specialists

6

Dashboard Tabs

8

UC Tables

0

Python Sidecars

◆ QUICK START GUIDE

1

Clone Repository

Clone databricks-industry-solutions/energy-sandbox and cd into production-optimizer/

2

Build Frontend + Backend

Run `npx tsc + cd ui && npm install && npx vite build`

3

Create Genie Space

Scope a Genie Space to the 8 gold tables and paste the Space ID into `app.yaml`

4

Deploy via DAB

`databricks bundle deploy --t prod` — Genie + Supervisor are live

◆ WHAT'S UNIQUE TO DATABRICKS

- **AI/BI Genie** over governed UC tables — NL → SQL with audit trail
- **Foundation Model API** for the decline analyst specialist + verdict synthesis

◆ TRY IN THE APP

- **Ask Genie** "Top 5 recommendations by annual_revenue" — see governed SQL execute
- **Approve a recommendation** via Supervisor — watch 5 specialists stream a verdict
- **Browse decline curves** in the Production Optimizer tab

- **UC SQL** via the Statement Execution API for 3 of 5 specialists
- **Unity Catalog** tags, lineage, and grants on every artefact
- **Databricks Apps** — one container, Node-native, no Python sidecar required

- **Walk the architecture** in the Data & AI Flow tab
- **Inspect the field** in 3D in the Digital Twin tab

◆ TARGET INDUSTRY

Upstream Oil & Gas — Production engineering and field development on conventional, unconventional, and EOR-flooded assets. Decline-curve analysis, recommendation approval, and self-service field insight built on the Databricks Lakehouse.

⚠ DEMO ONLY — This is a Databricks solution accelerator. See LICENSE for full terms of use.

© 2026 Databricks, Inc. · databricks.com · Databricks Industry Solutions